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Further palaeomagnetic results from the Sierras Australes fold and thrust belt, Argentina

R. N. Tomezzoli

Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Universidad de Buenos Aires, Instituto de Geofísica 'Daniel Valencio', Departamento de Ciencias Geológicas, Facultad de Ciencias Exactas y Naturales, Pabellón 2, *Ciudad Universitaria,* C142EHA, *Buenos Aires, Argentina*

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SUMMARY

The southwestern Gondwanaland margin during the late Palaeozoic was located between the South America craton and the Patagonia terrane. The Sierras Australes fold and thrust belt of Buenos Aires province (Argentina) has been interpreted as being the result of a crustal collision between these two blocks. A palaeomagnetic study of the less deformed upper part of the Permian Tunas Formation of the Pillahuincó region within the Sierras Australes belt was carried out in order to obtain reliable palaeomagnetic poles to constrain the palaeogeographic and tectonic evolution of this part of Gondwana. More than 300 specimens were measured from the Tunas Formation in the Sierra de Pillahuincó, from 27 sites in four localities of these red sandstones and clay-siltstones. Demagnetization at high temperatures isolated a reversed characteristic remanent magnetization, consistent with magnetization acquired during the Kiaman reverse superchron. Stepwise tectonic tilt correction suggests that the upper part of the Tunas Formation acquired its magnetization before or early in the deformation. The position of the palaeomagnetic pole on the apparent polar wander path of South America suggests that the magnetization was acquired some time between the late Early Permian and the early Late Permian. From the best grouping of the partially tilt-corrected remanence directions, at 90 per cent unfolding for 24 sites, a mean palaeomagnetic pole Tunas II for the Tunas Formation of the Pillahuincó region was calculated. The position is longitude 025.9°E, latitude 74.1°S; $A_{95}=5.2°$, $K=33.6$. This pole is consistent with previous poles from South America assigned to a similar age and allows for a more accurate definition of the apparent polar wander path. An earlier study in the lower part of the Tunas Formation (Tunas I) shows syntectonic magnetization acquired during the Early Permian. Therefore the differences in the type and age of magnetization between the lower and upper parts of the Tunas Formation suggest that the deformation was diachronous and propagated more slowly towards the foreland. Palaeogeographically, these two Tunas Formation poles are interpreted as recording a slight counterclockwise movement of South America during the Permian.

Key words: Gondwana, palaeomagnetism, Palaeozoic, Sierras Australes, South America, tectonics.

INTRODUCTION

The southwestern margin of Gondwanaland was located north of the Colorado River during the Palaeozoic, and marks the boundary between the South America craton to the north and the Patagonia region to the south (Fig. 1). The Sierras Australes orogenic belt trends approximately NW–SE in the southwestern part of Buenos Aires province (Figs 1 and 2), and is considered to be the South American counterpart of the Cape Fold Belt in South Africa (Du Toit 1927). Its geological features have attracted the attention of many geologists since the early 1900s because it is a key to understanding the controversial evolution of this part of Gondwana. For some authors, the Sierras Australes are the result of intracontinental deformation exclusively (Sellés Martínez 1989; Cobbold *et al.* 1991; Japas 1989); for others, the Sierras Australes are the result of the collision of the Patagonia terrane with Gondwana (Dalmayrac *et al.* 1980; Ramos 1984; Tomezzoli 1997). The deformation has traditionally been regarded as having occurred in a single phase during the Late Permian–Triassic interval, by

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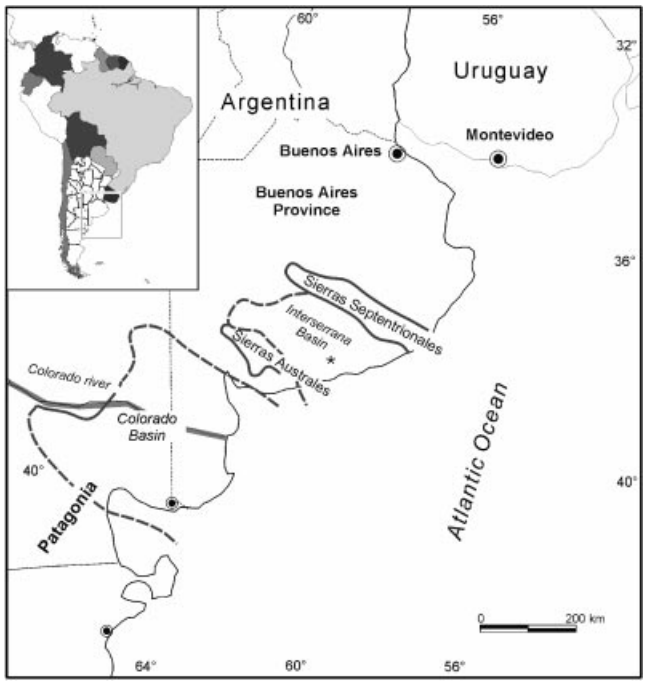


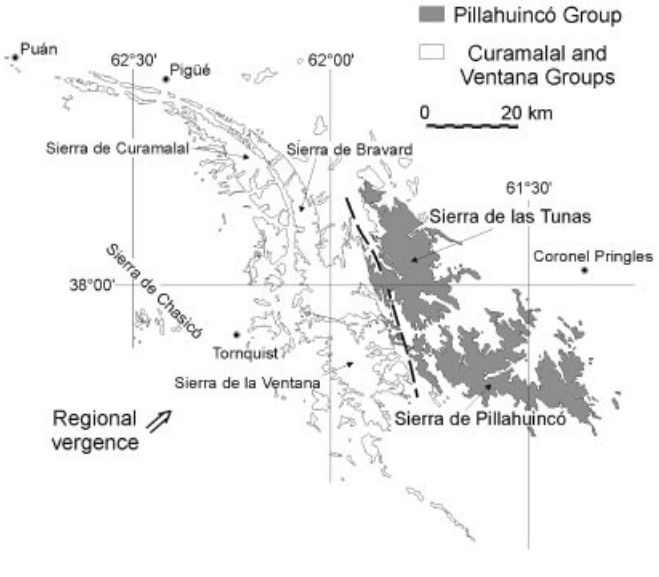
Figure 1. Location map of the Sierras Australes thrust and fold belt in the southwest Buenos Aires province, Argentina. The asterisk (*) indicates the González Cháves outcrops in the Interserrana basin.

analogy with the Cape Fold Belt (e.g. Harrington 1947; Japas 1989; Von Gosen *et al*. 1990; Rossello *et al*. 1997). Some investigators have suggested that there may have been more than one tectonic phase in this area, and postulated an early deformation phase of Late Devonian–Early Carboniferous age that preceded the widely recognized Upper Palaeozoic phase (Kilmurray 1975; Varela 1978; Ramos 1984, 1988; Tomezzoli 1997).

Tomezzoli & Vilas (1999) reported the presence of a syntectonic magnetization carried by haematite (at 42 per cent unfolding) in the red sandstone and clay siltstone of the lower part of the Tunas Formation in the Sierra de las Tunas region (Fig. 2). They concluded that the lower part of the Tunas Formation was deformed and remagnetized very soon after its deposition in the Early Permian. According to Ramos (1984), along the northern boundary of Patagonia and southern margin of Gondwana, an ocean was consumed by subduction beneath Patagonia, the subduction ceasing with the collision of the continental blocks. As a consequence of this collision, deformation in the Sierras Australes may have started in the mid-Palaeozoic. Available palaeomagnetic data for Patagonia (Rapalini & Vilas 1991; Rapalini *et al*. 1994) are consistent with the palaeolatitude expected from the South American poles for Devonian times onwards. The lack of significant latitudinal displacement of Patagonia with respect to Gondwana implies that accretion of Patagonia to Gondwana must have occurred during or before Devonian times, or else involved predominantly longitudinal convergence. The prolongation of the deformation until the Permian (Tomezzoli & Vilas 1999) may represent a continental tectonic readjustment. The main objective of the present palaeomagnetic study was to obtain additional reliable palaeomagnetic poles to evaluate the palaeogeographic and geodynamic evolution of this area, and therefore of Gondwana, building on the earlier study of Tomezzoli & Vilas (1999).

GEOLOGICAL SETTING

The Sierras Australes are a fold and thrust belt with a northeast vergence (in present geographical co-ordinates, Fig. 2). The range can be subdivided into two main orographic units (Harrington 1947; Fig. 2) with an important difference in the style of the deformation and grade of metamorphism, from



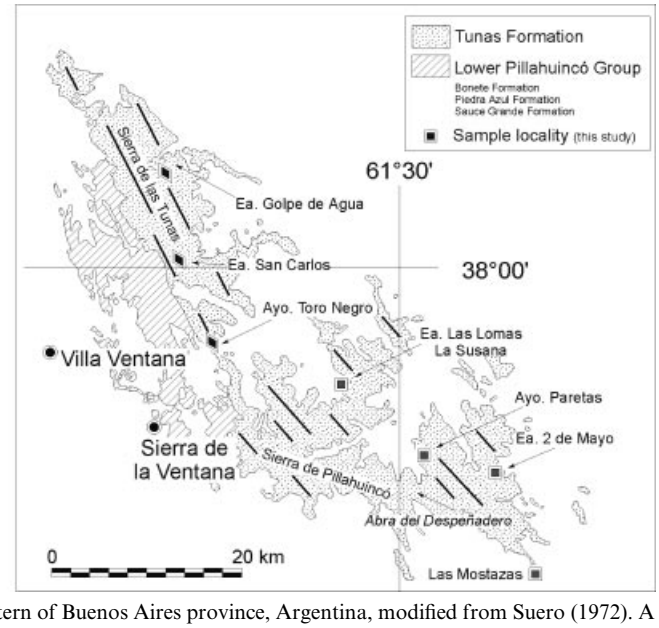


Figure 2. Geological map of the Sierras Australes exposed in the southwestern of Buenos Aires province, Argentina, modified from Suero (1972). A detailed map of the Tunas Formation shows the palaeomagnetic localities in the Sierras de las Tunas and in the Sierra de Pillahuincó (right). Diamonds are localities sampled by Tomezzoli & Vilas (1999). Squares are localities sampled in this study. The lines in the Tunas Formation indicate the average trend of the main structures. The dashed line on the left map divides the two orographic units: the highly deformed (on the left) Curamalal (Cambro-Ordovician) and Ventana Groups (Silurian–Devonian), and the less deformed Pillahuincó Group (Carboniferous–Permian).

lower greenschist in the Curamalal Group (Cobbold *et al*. 1986; Buggisch 1987) to diagenesis in the Pillahuincó Group. In the western sector are the highly deformed strata of the Sierras de Curamalal, Bravard and Ventana (Fig. 2). In the eastern sector are the Sierras de las Tunas and Pillahuincó (Fig. 2), with characteristically more open folding, which continues to the northeast in the subsurface (Llambías & Prozzi 1975), appearing again in the isolated small horizontal outcrop at González Chaves (Fig. 1).

Sampling was restricted to the Tunas Formation (Harrington 1947), which is widely exposed in the eastern sector of the Sierras Australes, both in the Sierra de las Tunas and in the Sierra de Pillahuincó. It is the youngest unit of the Carboniferous–Permian Pillahuincó Group (Furque 1973), which is composed of the Sauce Grande, Piedra Azul, Bonete and Tunas Formations (Fig. 2). The palaeoflora and palaeofauna associations of the Tunas Formation indicate an Early Permian to early Late Permian age for these rocks (Harrington 1947; Archangelsky & Cúneo 1984). This formation comprises over 1200 m of silicified fine pale green sandstone and lithic feldsarenite interbedded with red clay–siltstone and shale, which were part of a deltaic complex prograding towards the northeast (Harrington 1947; Andreis *et al*. 1979; Andreis & Cladera 1992; López Gamundi *et al*. 1995). Fossil leaves and tuffaceous intervals are present. A characteristic of the shale and fine sandstone is the presence of red sediments, which according to Andreis *et al*. (1979) contain haematite as identified both by X-ray diffractometry and optically. The Tunas Formation is differentiated from other formations on the basis of its different compositional suite, a well-defined northeastward palaeocurrent pattern, and different depositional conditions. The palaeocurrent data suggest a unimodal distribution with flow from the southwest towards the northeast (López Gamundi *et al*. 1995). The strata are older in the western areas (e.g. the San Carlos palaeomagnetic sampling locality shown in Fig. 2) and younger towards the east (e.g. Las Mostazas palaeomagnetic locality in Fig. 2). They are more deformed in the base of the formation (e.g. in San Carlos the folds are well defined) than in the top (e.g. in Las Mostazas where the sequence is quasi-horizontal).

LABORATORY PROCEDURES

Palaeomagnetic results from four localities (see Fig. 2 and Tables 1 and 2) are presented in this work: Ea. (Estancia, farm) Las Lomas-La Susana, Ayo. (Arroyo, stream) Paretas, Ea. Dos de Mayo (formerly San Miguel), and Cantera (quarry) Las Mostazas. Each locality is about 4 km^2 in areal extent and includes several sites located on different limbs of a folded sequence, representing different chronostratigraphic levels. At least four hand samples or six drill cores were collected at each of the 27 sites. They were orientated in the field using magnetic and/or sun compasses. Usually, three standard cylindrical specimens (2.2 cm long $\times$ 2.5 cm diameter) were cut from each core. Measurements of natural remanent magnetization (NRM) were made using a DC squid cryogenic magnetometer (2G model 750R) at the 'Daniel Valencio' palaeomagnetic laboratory of the Universidad de Buenos Aires. Pilot samples were subjected to stepwise alternating field (AF) demagnetization in 15 steps, up to 140 mT. This method proved effective in demagnetizing only some of the specimens from Las Mostazas, due to the high coercivities of the remanence in other localities. Thermal demagnetization was more successful, and was applied in at least 15 steps, with maximum temperatures of 680 to 730 °C, using a Schonstedt Instrument TSD-1 oven. Bulk susceptibility was measured after each step to monitor possible chemical changes during heating, using a RMSH-III TATA (Institute of Fundamental Research, Bombay, India) susceptibility meter.

Demagnetization results were analysed using orthogonal vector plots (Zijderveld 1967) and stereographic projections. Palaeomagnetic directions were determined using principal component analysis (Kirschvink 1980) or using the remagnetization plane technique (Halls 1976). The final site-mean directions were compiled using Fisher (1953) statistics, or in the case of combined directional data and remagnetization great circles, the method of McFadden & McElhinny (1988).

PALAEOMAGNETIC STUDY

Natural remanent magnetization intensities range between 0.5 and 90 mA m^{-1}. All samples exhibit similar behaviour during progressive thermal demagnetization. Most of them were quite stable during the early heating steps and started to demagnetize between 500 and 680 °C with a gradual quasi-linear decay towards the origin (Figs 3a and b). Other samples displayed an abrupt decay of a very stable remanence (Fig. 3c). Unblocking temperatures between 630 and 680 °C clearly show that this magnetization is carried by haematite. In some samples from Las Mostazas, the demagnetization paths were curved, along great circles towards the southeast quadrant with positive inclination, indicating simultaneous removal of distinct magnetic components with overlapping unblocking temperatures (Fig. 3d). Assuming this, the method of McFadden & McElhinny (1988) was used to analyse these cases. Unblocking temperatures of samples from the Las Mostazas locality range between 450 and 680 °C, showing that this magnetization is carried by magnetite, haematite or both. Further rock-magnetic studies are required to confirm the magneto-mineralogy. The difference between the magnetic behaviour of samples from this locality and the others is thought to be related to the different carriers of the magnetization, perhaps environmentally controlled in the depositional basin, recalling that the Las Mostazas locality corresponds to the upper level of the sampled units.

Table 1. Sample localities for the Tunas Formation study at the Sierra de Pillahuincó.

Locality	Latitude	Longitude	Sites	Specimens
Las Lomas-La Susana	38°8.9′S	61°34.4′W	SL1–SL7	106
Ayo. Paretas	38°1.5′S	61°29.2′W	AP1–AP5	40
Dos de Mayo	38°10′S	61°25.0′W	Y1–Y4; YD1–YD4	96
Cantera Las Mostazas	38°16′S	61°21.5′W	LM1–LM7	88

Table 2. Site-mean high-temperature characteristic directions of Population 1 in the Tunas Formation in the Sierra de Pillahuincó. *N/n*: number of processed specimens/number of specimens used in the calculation of the mean; (*n*) number of great circles. Dec.: declination (°); Inc.: inclination (°); α_{95} (°): semi-angle of the 95 per cent confidence cone; *k*: Fisher statistical parameter (Fisher 1953). B. Plane: bedding, strike and dip (90° cw from given strike). The mean at 90 per cent unfolding was used to calculate the Tunas II PP.

Site	*N/n*	Dec.°	Inc.°	α_{95}°	*k*	B. Plane	Dec.°	Inc.°	α_{95}°	*k*	Dec.°	Inc.°	α_{95}°	*k*
		in situ					with structural correction							
		Las Lomas-La Susana					**100 per cent**				**92 per cent**			
SL1	15/7(4)	106	73	12	29.5	135/27	184.5	66	12	29.5	181	68	12	29.5
SL2	11/9(2)	187	32.5	5	122	349/31	165	37	5	122	167	37	5	122
SL3	16/16	179.5	39	4	78.5	312/29	152.5	56	4	78.5	155	55	4	78.5
SL4	19/12(6)	186	25	6.5	48	336/42	159.5	38	6.5	48	162	38	6.5	48
SL5	19/19	150	57	5	40.5	150/18	176	53	5	40.5	174	53	5	40.5
SL6	14/13	123	62	5	63	158/19	163	67	5	63	159	67	5	63
SL7	12/9(5)	133	56	7	66	158/19	163.5	59	7	66	161.5	59.5	7	66
Mean	7/7	162.6	52.7	19.8	10.2		165.3	54.2	10.2	35.8	162.2	54.2	10.1	36.4
		Arroyo Paretas					**100 per cent**							
AP2	11/11	122	47.5	5	92	148/28	157	53	5	92	–	–	–	–
AP3	8/8	116	43	5	143	155/30	149	55	5	143	–	–	–	–
AP4	6/6	107	48	7	86	145/32	149.5	58	7	86	–	–	–	–
AP5	6/6	110.5	64	5	218	142/41	189.5	56	5	218	–	–	–	–
Mean	4/5	114.0	50.5	11.5	65	–	160.7	56.5	12	55.5	–	–	–	–
		Dos de Mayo					**100 per cent**				**85 per cent**			
Y1	13/9	86	64	5	97	135/27	156	72	5	97	144	73	5	97
Y2	12/12	86	58	5	81.5	136/42	174.5	64	5	81.5	162	68	5	81.5
Y3	12/12	103	615	3.5	151	136/42	181	56	3.5	151	173	60	3.5	151
Y4	10/10	93.5	47	7	49	138/41	151	60	7	49	141	61	7	49
YD2	12/9	189	43	7	60	313/37	145	65	7	60	156	63.5	7	60
YD3	11/11(3)	190	49	9	27	318/30	148	65.5	9	27	157	64.5	9	27
YD4	14/11(4)	178	41	9	27	311/33	143	58.5	9	27	150.5	57	9	27
Mean	7/8	139.1	61.8	23.9	7.3	–	157.4	63.8	6.6	84.4	155.1	64.2	5.4	126.2
		Las Mostazas					**100 per cent**							
LM1	12/9(3)	158	47	12	20	240/7	159	54	12	20	–	–	–	–
LM2	9/9(3)	113	66	9	35	118/29	163	54	9	35	–	–	–	–
LM3	16/15(13)	155.5	25	7.5	29	148.5/5	158	24.5	7.5	29	–	–	–	
LM4	12/8(8)	157.5	48	10	26	134.5/12	168	42	10	26	–	–	–	–
LM5	13/8(4)	160	41	12	23	138/9.5	167	37	12	23	–	–	–	–
LM6	15/15	140	50	4	77	154.5/10	152	51	4	77	–	–	–	–
Mean	6/7	150.6	47.0	14	23.5	–	161.4	44.0	10.5	42	–	–	–	–
		Sierra de Pillahuincó					**100 per cent**				**90 per cent**			
Mean	24/27	144.9	54.6	9.4	10.9	–	161.6	54.9	4.9	37.6	160.1	55.2	4.8	38.7

The high-temperature characteristic remanent magnetization (ChRM) is referred to as 'Population 1' (Table 2), for the different localities. It always shows positive inclinations (downward directed vectors) and good within-site groupings ($\alpha_{95} < 15°$ and $k > 20$), except for three sites, namely AP1, YD1 and LM7, where the site-mean ChRM was poorly determined with α_{95} greater than 15°. For this reason, palaeomagnetic data from these three sites were rejected. In approximately 30 per cent of the specimens, a second low-temperature component called 'Population 2', was isolated between 100 and 450 °C (Figs 3a and b). It has negative inclinations (*in situ*), with northward declinations (close to 0°) and a tendency to be more scattered. The low unblocking temperatures and the present-day field direction indicate that the Population 2 component is a secondary magnetization of recent vintage. Consequently, it is not considered relevant to the tectonic objective of this paper.

Progressive unfolding was applied to the Population 1 vectors from Las Lomas-La Susana (Table 2). The *in situ* mean direction (Fig. 4a1) is decl. = 162.6°, incl. = 52.7°, α_{95} = 19.8° and k = 10.2. After full bedding correction the mean direction is decl. = 165.3°, incl. = 54.2°, α_{95} = 10.2° and k = 35.8 (Fig. 4a2). Application of a stepwise structural correction shows a clear improvement in grouping of directions at around 92 per cent unfolding (Fig. 4a3): decl. = 162.2°, incl. = 54.2°, α_{95} = 10.1° and k = 36.4. A minimum fold test statistic (SCOS) of McFadden 1990 was observed at 83 per cent. Using the Watson & Enkin (1993) fold test, the optimum degree of unfolding occurs at 90.6 per cent, with 2.5 and 97.5 percentile limits at 77.7 and 105.9 per cent, respectively. This result suggests that the 100 per cent unfolding case cannot be excluded. Therefore, the magnetization of the Tunas Formation in Las Lomas-La Susana was acquired before or during the initial stages of folding of the sequence. The lower temperature Population 2 *in situ* mean direction is decl. = 17.0°, incl. = −58.0°, α95 = 17° and k = 14. This direction is not significantly different from the expected geocentric axial dipole field direction (decl. = 0°, incl. = − 57.4°), and therefore is interpreted as a secondary magnetization acquired during recent times.

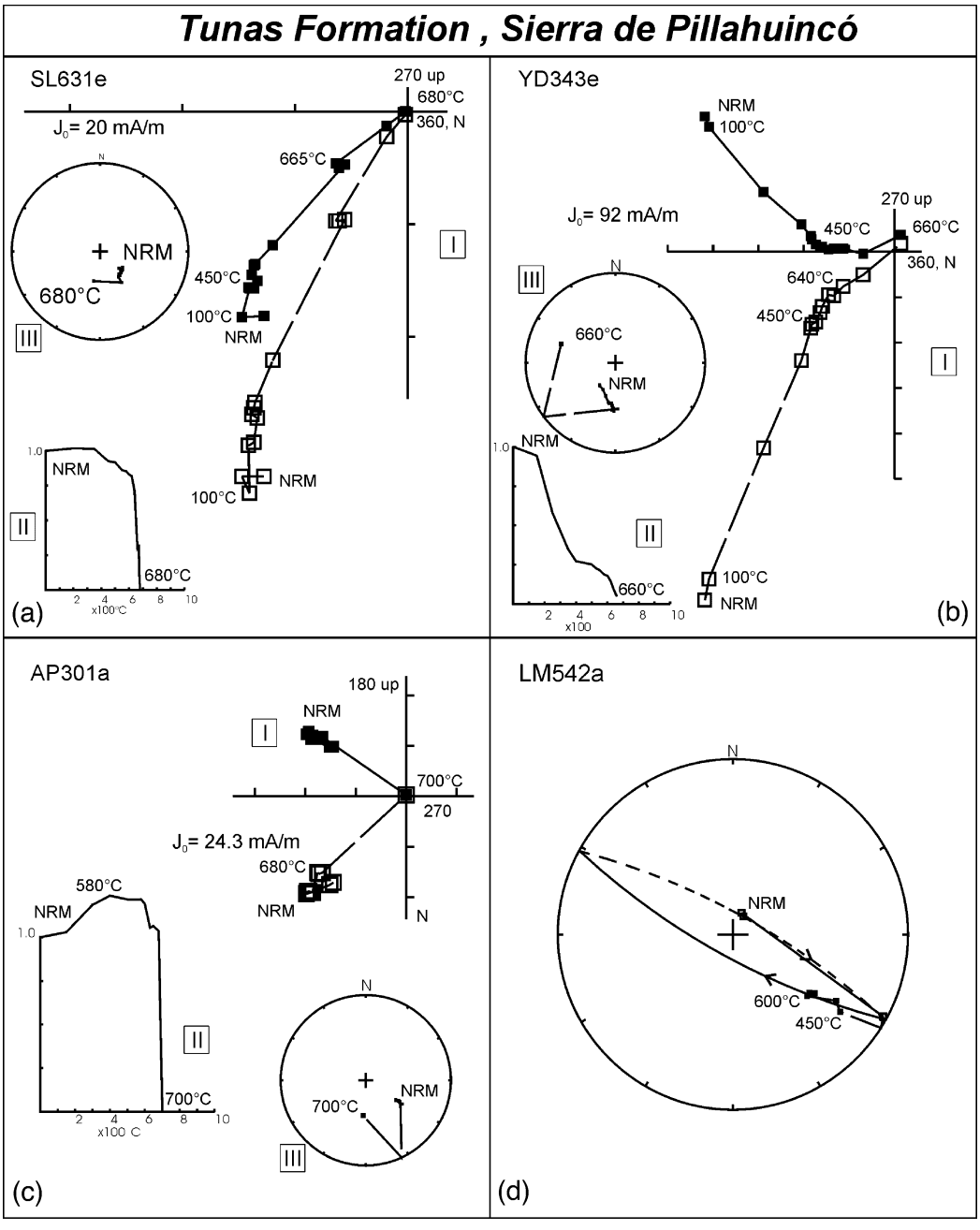


Figure 3. Representative thermal demagnetization results from some specimens from different localities (Fig. 1). I: Zijderveld diagrams. Open (filled) squares indicate representation in the vertical (horizontal) plane, in geographical coordinates. II: demagnetization curves. III: equal-area stereographic projection. In **(d)**, a remagnetization circle is shown.

In Arroyo Paretas the *in situ* group mean direction of Population 1 (Fig. 4b1), with $N=4$ is decl. $=114.0°$, incl. $=50.5°$, $\alpha_{95}=11.5°$ and $k=65$. Full bedding correction yields a mean direction of decl. $=160.7°$, incl. $=56.5°$, $\alpha_{95}=12°$ and $k=55.5$ (Fig. 4b2), increasing the dispersion slightly ($k=65$ versus $k=55.5$). With the Watson & Enkin (1993) fold test, the optimum degree of unfolding occurs at 22.5 per cent, with 2.5 and 97.5 percentile limits occurring at -39.9 and 93.1 per cent, respectively. This large uncertainty arises from the slight differences in bedding attitudes. Thus, the conclusion could be (1) a syn-folding magnetization or (2) a post-folding magnetization. However, from examining the distribution of site-mean directions (Fig. 4b) it can be clearly seen that the dispersion lies in the declination components, which could originate from a minor local rotation. In this case, inclination-only statistics would be better for evaluating the relative timing of the magnetization. The McFadden & Reid (1982) and the Watson & Enkin (1993) fold tests based on inclination-only statistics both show that the optimum degree of unfolding occurs at 100 per cent unfolding (error estimates for the Watson and Enkin fold test are 76.4 per cent at 2.5, and 131.4 per cent at 97.5). Thus the Ayo. Paretas magnetization cannot be post-folding, and it is most probably pre-folding. The low blocking temperature Population 2 retained a highly scattered distribution both *in situ* and after

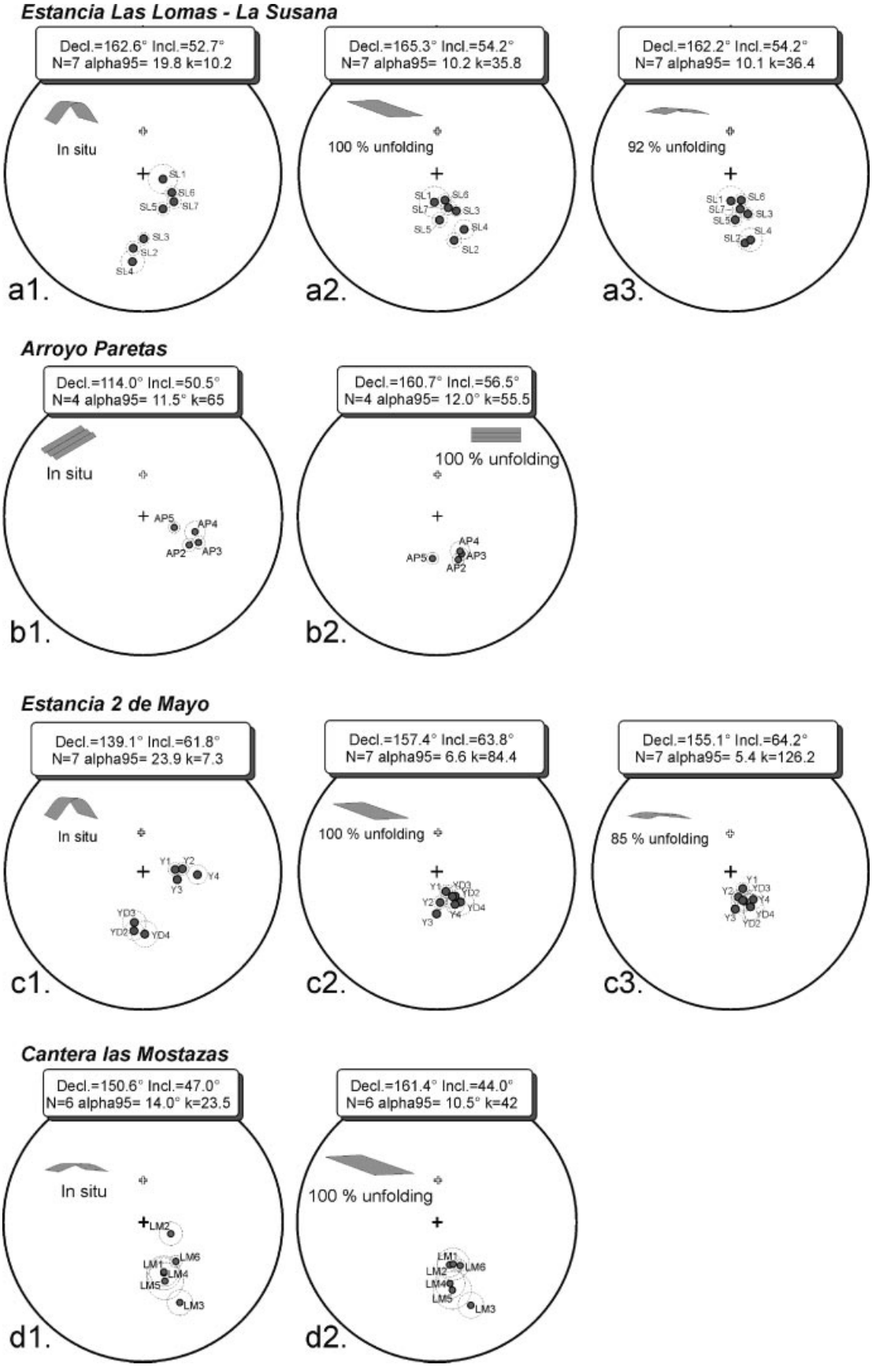


Figure 4. Tectonic tilt corrections for Population 1 (high-temperature characteristic remanence) of the Tunas Formation in (a) Estancia Las Lomas-La Susana; (b) Arroyo Paretas; (c) Estancia Dos de Mayo; and (d) Cantera Las Mostazas. (1) *In situ*; (2) fully (100 per cent); and (3) partially tilt-corrected. See also Table 2. The axial dipole field direction is indicated.

structural correction. At only one site is this component *in situ* close to the geocentric axial dipole field direction, such that it appears to be a secondary magnetization of recent origin.

The *in situ* group mean direction of Population 1 (Fig. 4c1) in Ea. Dos de Mayo (Fig. 2) is decl. = 139.1°, incl. = 61.8°, $\alpha_{95} = 23.9°$ and $k = 7.3$, $N = 7$. Full bedding correction decreases the dispersion: decl. = 157.4°, incl. = 63.8°, $\alpha_{95} = 6.6°$ and $k = 84.4$ (Fig. 4c2). Application of stepwise structural correction shows the best grouping of directions (Fig. 4c3) near 85 per cent unfolding: decl. = 155.1°, incl. = 64.2°, $\alpha_{95} = 5.4°$ and $k = 126.2$. The SCOS parameter correlation (McFadden 1990) was at 87 per cent bedding correction. According to the McElhinny (1964) method, in comparing these precision parameters before and after the bedding correction this magnetization is pre-folding at the 95 per cent confidence level. For this locality, application of the Watson & Enkin (1993) fold test excludes the 100 per cent case (83.9 per cent optimum, with 75.5 and 93.7 per cent at 2.5 and 97.5 limits). These data suggest that the magnetization of the Tunas Formation at the Ea. Dos de Mayo locality was acquired at an early stage in the folding of the sequence. The *in situ* lower-temperature component mean direction is decl. = 327.0°, incl. = −47.4°, $\alpha_{95} = 27°$ and $k = 22.5$, $N = 3$. This direction includes the geocentric axial dipole field direction and is interpreted as a secondary magnetization acquired during recent times.

In Cantera Las Mostazas (Table 2) the *in situ* mean direction of Population 1 (Fig. 4d1) is decl. = 150.6°, incl. = 47.0°, $\alpha_{95} = 14.0°$ and $k = 23.5$. After full bedding correction the mean direction is decl. = 161.4°, incl. = 44.0°, $\alpha_{95} = 10.5°$ and $k = 42$ (Fig. 4d2). Application of the tilt correction shows an improvement of grouping of directions at 100 per cent unfolding from $k = 23.5$ to $k = 42$ (Fig. 4d2). The Watson & Enkin (1993) fold test using 1000 simulations and a 5-degree uncertainty on bedding shows that the optimum degree of unfolding occurs at 95.3 per cent, with 2.5 and 97.5 uncertainty limits occurring at 46.7 and 139.1 per cent unfolding, respectively. This fold test does not exclude the 100 per cent unfolding case. The low-temperature component mean direction grouped as Population 2 is decl. = 347.2°, incl. = −58.5°, $\alpha_{95} = 6.4°$, $k = 91$ and $N = 7$, similar to the geocentric axial dipole field direction, and is interpreted as a secondary magnetization acquired during recent times.

In summary, the overall mean direction based on all accepted sites ($N = 24$) for the Tunas Formation in the Sierra de Pillahuincó is *in situ*: decl. = 144.9°, incl. = 54.6°, $\alpha_{95} = 9.4°$ and $k = 10.9$. The 100 per cent bedding-corrected mean is decl. = 161.6°, incl. = 54.9°, $\alpha_{95} = 4.9°$ and $k = 37.6$. The best grouped mean occurs at 90 per cent of tilt correction: decl. = 160.1°, incl. = 55.2°, $\alpha_{95} = 4.8°$ and $k = 38.7$, consistent with a pre-tectonic magnetization at the 99 per cent confidence level (Fig. 5) according to the correlation parameters of McFadden (1990). A minimum fold test statistic (SCOS) of McFadden 1990 was observed at 91 per cent. Using the Watson & Enkin (1993) fold test the optimum degree of unfolding occurs at 87.5 per cent, with 2.5 and 97.5 uncertainty limits occurring at 80.5 and 94.6 per cent unfolding, respectively.

DISCUSSION

Results presented here support the interpretation that the rocks of the upper part of the Tunas Formation in the Sierra de Pillahuincó have a characteristic stable remanent magnetization with exclusively reverse polarity, consistent with the acquisition of magnetization during the Kiaman reverse polarity superchron. The carrier of the magnetization of these rocks is mainly haematite. Andreis *et al.* (1979) have identified the presence of haematite both petrographically and by X-ray studies. Typically the rocks are reddish in colour. Furthermore, the high blocking temperatures indicate that the magnetization is stable up to the Curie temperature of haematite. The magnetization could be carried either by pigment grains, detrital grains or both. However, the characteristic directions isolated by Zijderveld analyses indicate a single high blocking-temperature component. Because these rocks have not been significantly metamorphosed, but have suffered only diagenetic changes (Cobbold *et al.* 1986 and Buggisch 1987), the magnetization is believed to be mainly syn-depositional to syn-diagenetic.

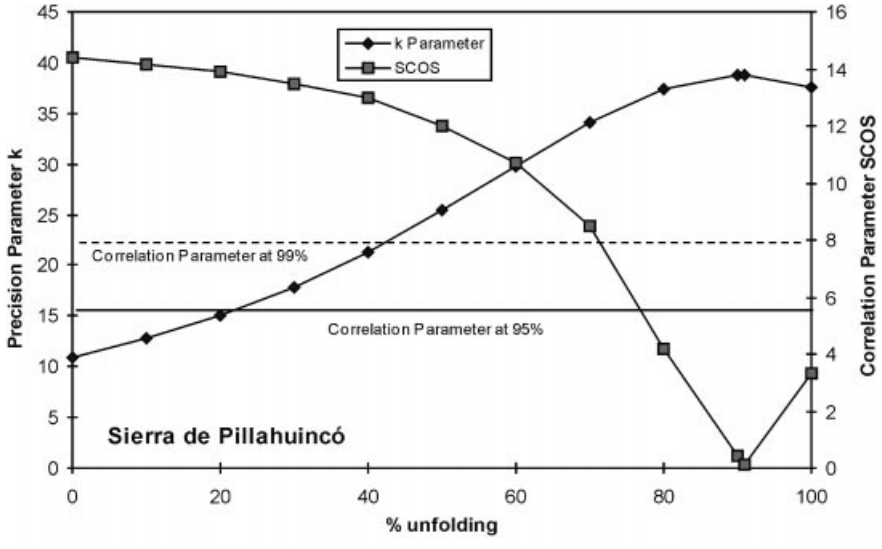


Figure 5. Plot of the statistical parameter k (Fisher's precision parameter) and ξ (SCOS: correlation parameter of McFadden 1990), for the site-mean directions (four localities, 24 sites) of the Tunas Formation from the Sierra de Pillahuincó, versus percentage of tectonic correction. Lines are significance levels at 99 per cent and 95 per cent according to McFadden (1990). See also Table 2.

The resulting palaeomagnetic pole (PP) is called 'Tunas II' (Pillahuincó), and was computed from virtual geomagnetic poles (VGP) representing the four sample localities. To obtain the VGPs, the mean directions for these four localities were jointly treated to an incremental unfolding test. The best grouping of these vectors was obtained for 90 per cent unfolding. The VGPs corresponding to these vectors at 90 per cent unfolding were averaged to obtain the Tunas II pole at latitude 74.1°S; longitude 25.9°E, $A_{95}=5.2°$ and $K=33.6$, $N=24$ (Table 3). However the PP is not significantly different using the 100 per cent unfolding direction, namely latitude 75.2°S; longitude 28.3°E, $A_{95}=5.3°$ and $K=31.7$, $N=24$. Using the VGPs calculated for each accepted site, based on the appropriate degree of unfolding from the field tests that yield the maximum grouping of the final mean direction (McFadden 1990), the PP calculated is latitude 74.8°S; longitude 27.0°E, $A_{95}=5.2°$ and $K=33.4$, $N=24$.

Palaeomagnetic studies of the lower part of the Tunas Formation (Tomezzoli 1997; Tomezzoli & Vilas 1999) in the area of the Sierra de las Tunas (San Carlos, Golpe de Agua and Toro Negro localities; see Fig. 2) led us to conclude that these rocks are carriers of a syntectonic magnetization acquired during the folding of the sequence at 42 per cent unfolding. The depositional environment for the Tunas Formation progressed from a marine platform to a continental facies (Zavala *et al*. 1993) with sediment and tectonic transportation to the northeast (Cobbold *et al*. 1991; López Gamundi *et al*. 1995). The most important differences between the localities in the lower part of the Tunas Formation (in Sierra de las Tunas) and those from the upper part of the Tunas Formation (in the Sierra de Pillahuincó) are the intensity of the deformation and their position in the stratigraphic column. The older deposits of the Tunas Formation (at San Carlos, Fig. 2) are more strongly folded than the younger subhorizontal outcrops (at Las Mostazas, Fig. 2). This tectonic pattern is consistent with the regional northeast vergence of the Sierras Australes. In terms of magnetic characteristic behaviour, the rocks of the lower and upper Tunas Formation have acquired different types of magnetization. The magnetization is synfolding close to the fold and thrust belt (lower part of the Tunas Formation) and pre-folding near the foreland margin of the belt (upper part of the Tunas Formation). A syntectonic palaeomagnetic pole called Tunas I was calculated by Tomezzoli & Vilas (1999) for the localities in the Sierra de las Tunas (Fig. 2), at latitude 63.0°S; longitude 13.9°E, $A_{95}=5.4°$ and $K=39.7$, $N=19$. The Tunas I pole is concordant with other Early Permian poles in the APWP of South America (Fig. 6). The average palaeomagnetic pole for Tunas Formation calculated from all accepted sites (Tomezzoli & Vilas 1999 and this paper) is latitude 70.5°S; longitude 18.5°E, $A_{95}=3.5°$ and $K=27.2$, $N=44$.

In accordance with the geological evidence, the sedimentation was partially coeval with deformation (Cobbold *et al*. 1991), and the deformation is interpreted to have advanced from the west to the east. Thus, this pattern of magnetization and folding indicates that the rocks were remagnetized during a relative short period of time, as folding propagated more slowly towards the eastern foreland. In accordance with this interpretation, a virtual palaeomagnetic pole of possible Triassic age was calculated by Tomezzoli & Vilas (1997) for the isolated outcrop at González Chaves in the Interserrana Basin (see Figs 1 and 6). These rocks are correlated with the Pillahuincó Group (Llambías & Prozzi 1975). They lie horizontally in the depocentre of the basin and are composed of fine sandstone and clay silstone. On a regional scale, López Gamundi *et al*. (1995) interpreted this deformation and the coeval foreland basin phase as affecting diachronously the entire Panthalassan margin, ranging from the Early Permian in western Argentina, to slightly later in South Africa, to Late Permian in eastern Australia (Veevers *et al*. 1994). A similar case was identified by Stamatakos *et al*. (1996) in the Central Appalachians (USA), who found post-folding magnetizations near the fold and thrust belt, synfolding magnetizations in the central part of the belt, and pre-folding magnetizations near the foreland.

The Early Permian deformation responsible for the magnetization described in the Sierra de las Tunas (Tomezzoli & Vilas 1999) may correspond to the San Rafaelic orogenic phase (Azcuy & Caminos 1987; Ramos 1988). This phase has been recognized mainly in the western areas of Argentina (Llambías & Sato 1995), and has also been linked to regional-scale remagnetizations (Rapalini & Tarling 1993).

The Tunas II pole is compared with the late Palaeozoic APWP of South America in Fig. 6 and Table 3, and is thought

Table 3. Late Palaeozoic and Early Mesozoic South American poles selected from the cratonic areas of South America. PPs were taken from Van der Voo (1993) and McElhinny & Lock's (1996) Global Palaeomagnetic Database, and the data were checked with the original contributions. $A_{95}(°)$ = semi-angle of the 95 per cent confidence cone. The PP were selected based on a minimum of three of the reliability criteria of Van der Voo (1990); see Fig. 5.

Palaeomagnetic pole	Site			Pole			
	$A_{95}°$	Lat. °	Long. °	Plat. °	Plong. °	Age	References
González Chaves	17	−38	300	−84	216	248.5-Tr	Tomezzoli & Vilas 1997
Amana Ar.	8	−30.3	292.3	−83	317	248.5-lTr	Valencio *et al.* (1977)
Tunas II Ar.	5.2	−38.2	298.6	−74.1	25.9	275-ulP	This work
Itarare I2 Br.	4	−21.5	312.8	−60.3	29.5	290-lP	Pascholati (1983)
Tunas I Ar.	5.1	−38	298.2	−63	13.9	290-lP	Tomezzoli & Vilas 1999
Lw. Colorados Ar.	5	−29.5	293	−60	358	293-lP	Embleton (1970)
La Colina Ar.	8	−30	293	−49	343	294-lP	Sinito *et al.* (1979)
Itarare I Br.	4	−21.5	312.8	−56.7	350.6	295-lP	Pascholati (1983)
Tepuel Ar.	8.5	−43.5	289.6	−31.7	316	318-uC	Rapalini *et al.* 1994

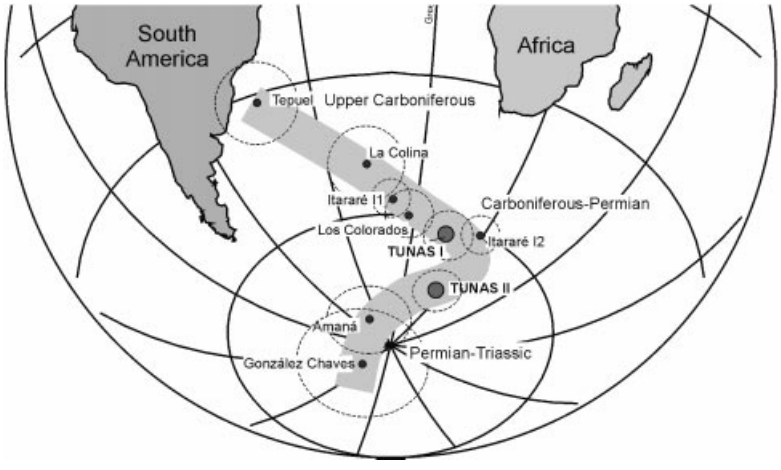


Figure 6. Position of the Tunas II (this study) palaeomagnetic poles computed from maximum grouping of the final mean direction at 90 per cent unfolding. Other Late Palaeozoic and Early Mesozoic South American PPs from the cratonic areas are also shown. These PPs were taken from Van der Voo (1993) and The Global Palaeomagnetic Database (McElhinny & Lock 1996). These PPs were selected based on a minimum of three of the reliability criteria of Van der Voo (1990). See also Table 3.

to correspond to the late Early Permian–early Late Permian. Its position on the APWP suggests a magnetization age younger than the Tunas I palaeomagnetic pole, consistent with their stratigraphic relations.

A possibility that also needs to be examined is that the declination difference of about 13° between the Tunas I and Tunas II PPs is due to local rotation during deformation. It is possible that the Tunas II region (i.e. in the Sierra de Pillahuincó) has rotated 13° relative to Tunas I. The trends of the major folds in both Sierra de las Tunas and Sierra de Pillahuincó are similar, but according to the regional maps (Suero 1972; Cobbold *et al*. 1991) the average azimuth of the structures in the Sierra de Pillahuincó (see Fig. 2) is a few degrees less (this is the only geological feature that can be interpreted as evidence of rotation in the area). However, if this difference were caused by rotation between blocks, it would require a counterclockwise rotation of the Sierra de Pillahuincó (Tunas II) relative to the Sierra de las Tunas (Tunas I), and this is opposite to the rotation required to explain the separation of the two palaeomagnetic poles. Block rotations could be expected in a transpressive context as some authors suggest. However, no evidence for such rotations has been identified in the Tunas Formation, and, furthermore, the sense of the movements in a regional view of the Sierras Australes has not been resolved, since there are different structural interpretations. Cobbold *et al*. (1991) and Japas (1989) view the Sierras Australes as a right-lateral transpressive fold belt; Sellés Martinez (1989) interpreted left-lateral wrench displacement in an E–W direction. Von Gosen *et al*. (1991) recognized sinistral and dextral shear planes, but concluded that the Sierras Australes are bounded by sinistral shear zones.

The different positions of the Tunas I and Tunas II poles (Fig. 6) can be interpreted as due to a change of orientation of the continent relative to the geographical south pole, with a small counterclockwise rotation between Early Permian and Late Permian times (Fig. 7). The declination difference for the Tunas I and Tunas II poles from the sampling region is $12.7° \pm 6.7°$. Each pole represents a significant stratigraphic

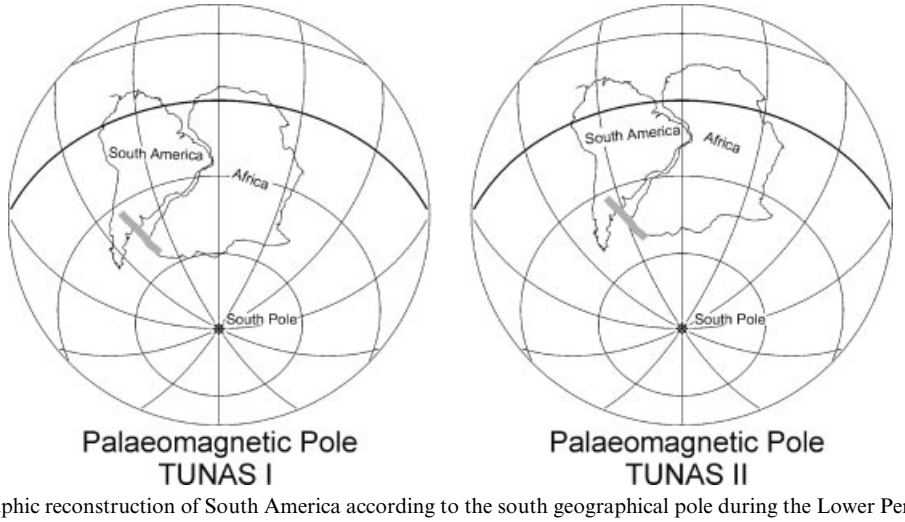


Figure 7. Palaeogeographic reconstruction of South America according to the south geographical pole during the Lower Permian, from the Tunas I and Tunas II poles. The Gondwana boundary is indicated in grey.

thickness, more than 500 m, therefore, because of the stratigraphic separation between Tunas I and Tunas II and because Tunas I is a syntectonic magnetization and Tunas II is a pretectonic magnetization, the age difference between the two poles is too great for the declination difference to be attributed to secular variation. The difference in the declinations might be attributed to apparent polar wander. The difference in latitudes of the sites relative to the two poles is $4.1^\circ \pm 5.3^\circ$, which is not significant at the 95 per cent confidence level.

CONCLUSIONS

A palaeomagnetic study of the upper part of the Tunas Formation in the Sierras Australes of Buenos Aires province shows that these rocks from Las Lomas–La Susana, Ayo. Paretas, Dos de Mayo and Las Mostazas (situated in the Sierra de Pillahuincó) localities are carriers of pre-tectonic magnetization. The palaeomagnetic pole defined from the best grouping mean direction at 90 per cent unfolding is latitude 74.1°S; longitude 25.9°E, $A_{95} = 5.2^\circ$ and $K = 33.6$, $N = 24$. However, the PP would not be different using the 100 per cent unfolding direction: latitude 75.2°S; longitude 28.3°E, $A_{95} = 5.3^\circ$ and $K = 31.7$, $N = 24$, or using the directions based on the optimum degree of unfolding at each individual locality: latitude 74.8°S; longitude 27.0°E, $A_{95} = 5.2^\circ$ and $K = 33.4$, $N = 24$.

This new Tunas II pole lies on the South American APWP. It is possible to constrain the age of the Tunas II (Pillahuincó) magnetization and thus of its palaeomagnetic pole in the late Early Permian–early Late Permian. The entire Tunas Formation has a distinctive temporal variation in the age of folding relative to remagnetization. Remagnetization is synfolding in the western area of the Tunas depositional basin (Tunas I PP; Tomezzoli & Vilas 1999) near the fold and thrust belt, and prefolding near the foreland basin towards the eastern area. This pattern of remagnetization and folding indicates that the rocks were remagnetized and folded during a relatively short period of time as folding propagated more slowly towards the foreland. The palaeogeographic implication of these data for South America (Tunas I and Tunas II poles) is consistent with a counterclockwise movement of the continent between the Early and Late Permian. Assuming that the collision between Patagonia and the South American craton began during the Devonian and continued until the Permian, this movement could be the consequence of a late adjustment between these continental blocks before the final assembly.

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